Tyler Chisholm

Technical Designer

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Technical designer who designs a system and builds it: gameplay C++ and Blueprint in UE5, C# in Unity, editor tools for the designers who tune it, and the docs that keep a team aligned. Led design on a UE5 capstone headed to Steam; deep rendering background.

Education

M.S. Interactive Entertainment, Florida Interactive Entertainment Academy (FIEA), University of Central Florida. Dec 2026
12 games built across 4 semesters in UE5, Unity and VR.

B.B.A. Finance & Entrepreneurship, Stetson University, DeLand, FL May 2024

Skills

Design	Systems and mechanics design, GDDs and one-page specs, rapid prototyping, economy and balance modeling, playtest analysis
Engines	Unreal Engine 5 (C++, Blueprint, Chaos, World Partition), Unity (C#, URP), Godot
Languages	C++, C#, Blueprint, HLSL, Python, Rust
Tools	Perforce, Git, Jira, Visual Studio, Rider, Blender, Nsight Graphics, Excel/VBA

Experience

Design Lead — Everent (Studio Kismet, FIEA capstone) UE5.7 · C++ · Blueprint · Chaos · Perforce Oct 2025 – Aug 2026

Momentum-based hoverboard game across an open world of floating sky islands, coming to Steam. Led 7 designers on a 20+ person team from concept through production; wrote the GDD and one-page specs; ran sprint planning and Jira for the design team.

- Designed and built speed-gated destruction in C++ on Chaos Geometry Collections: a predictive sweep measures impact speed into the wall, drops blockers a frame early to keep momentum, and fires strain, directional and radial velocity fields. It doubles as secret gating and skill expression.
- Built the Destructible Wall Browser editor tool: lists every destructible in the level, exposes its tuning and fires a test projectile at a set point and speed, so designers tune without entering Play mode.
- Implemented a C++ corrupt-surface physical material (world-space gravity, angle-scaled repulsion, boost suppression) as the constraint that keeps puzzles solvable in a game where the player can ride any surface.

Project Lead — Studio Microdose (FIEA Ventures) Multiplayer co-op · publisher-pitch vertical slice Aug 2026 – Present

- Leading a 10-person team on a co-op game set in the Florida Everglades where players catch invasive animals by hand; hands-on in the build while running scope and leads (art, design, tech, community).

Selected Projects

Spatial Engine — path-traced VR engine Rust · Vulkan RT · HLSL · OpenXR 2026 – Present

Personal engine: architected and directed; implementation AI-assisted.

- Runs the full path tracer in VR at 90 Hz on a Quest over Link on modest RTX hardware, across ~900 km of streamed terrain with volumetric clouds.
- Physics-driven bodies: a Jolt active-ragdoll player with a contact-based support model for walking and climbing, and powered-ragdoll NPCs whose wound list (stab, stick, carve, fractures) drives bleeding and knockouts.

Pocket Spaces — stencil portal system (bigger on the inside) Unity URP · C# 2026

- Doorways reveal rooms larger than their exteriors via stencil masking; a player state machine, per-object portal membership and gravity overrides let the player and physics objects cross cleanly.

Star Fox Adventures on PS Vita — native port C · GXM · PS Vita · released 2026

- Native PS Vita port built on the community SFA decomp: the decompiled game runs on the Vita's ARM CPU, and I wrote the graphics backend that translates the GameCube's GX API to GXM.

Earlier Experience

Quality Assurance Agent — Stormbreaker Studios, Orlando, FL Apr – Jul 2020

Tested 8 iOS client apps; wrote 85 defect tickets with repro steps and fix recommendations.